Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Electronics Devices and Circuits	3	0	0	3	3
	PC	Electronics Devices and Circuits Laboratory	0	0	2	2	1
	PC	Networks and Systems	3	1	0	4	4
	BSC	Math-3	3	1	0	4	4
			15	**2**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Analog Electronics	3	0	0	3	3
	PC	Analog Electronics Laboratory	0	0	2	2	1
	PC	Analog Communication	3	0	0	3	3
	PC	Analog Communication Laboratory	0	0	2	2	1
	PC	Electromagnetics and Transmission Lines	3	1	0	4	4
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3

1st Semester

<Course Code>					Mathematics – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To be able to evaluate problems related to differential and integral calculus of complex functions.
2. To be able to obtain area, volume using integral calculus.
3. To be able to formulate and solve various engineering problems using the calculus.
4. To study the properties of sequence and series and to check the convergence and divergence.

UNIT I: DIFFERENTIAL CALCULUS AND ITS APPLICATIONS **08 Hrs.**

Partial derivative and its application - Euler's theorem - Total derivatives - Jacobians – Maxima and Minima of two variables using Lagrange's multipliers. Taylor's series, Convergence of power series.

UNIT II: INTEGRAL CALCULUS AND ITS APPLICATIONS **12 Hrs.**

Definition and evaluation of double integral (Cartesian – Polar form) – Change of orders - Change of variables – Evaluation of triple integral, change of variables (Cartesian to spherical – and cylindrical) – Applications, area – volume – center of mass – center of gravity by double and triple integral.

UNIT III: VECTOR CALCULUS **10 Hrs.**

Gradient, divergence and curl – Directional derivative – Irrotational and Solenoidal vector fields – Vector Integration – Simple problems on line, surface and volume integrals – Green's theorem in a plane, Gauss divergence theorem and Stokes' theorem (without proofs) – Simple application involving cubes and rectangular parallelepipeds.

UNIT IV: SEQUENCE AND SERIES **12 Hrs.**

Definition. Convergent, divergent, bounded & monotone sequences. Infinite sums: Basics Taylor series, Convergence and divergence, Tests for convergence for positive term series, Alternating series – Leibnitz test, Absolute convergence, conditional convergence.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify the use of convergence of infinite series in engineering aspects.
CO2 : Understand the concept of Directional derivative, Irrotational and Solenoidal vector fields.
CO3 : Apply the concept of differential and integral calculus in engineering problems.
CO4 : Analyze the obtained solution in linear and non-linear domains.
CO5 : Appraise mathematical problems from complex domain.
CO6 : Evaluate problems on Green's, Stokes' and Divergence theorems.

TEXT/REFERENCE BOOKS

1. B. S Grewal, Higher Engineering Mathematics, Khanna Pub.
2. R. K. Jain and S. R. K. Iyengar, Advanced Engineering Mathematics, Alpha Science.
3. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley.
4. G. Strang, Linear Algebra and its Applications, Cengage Learning.
5. K. Hoffman and R. A. Kunze, Linear Algebra, Prentice Hall of India.

<Course Code>					Applied Physics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand Electric and Magnetic Fields, applying vector algebra.
2. Apply Maxwell's equations to analyse electromagnetic waves and transmission.
3. Analyse semiconductor behaviour and its applications in electronic devices
4. Evaluate optical phenomena and their engineering applications in communications.

UNIT I: ELECTRICITY AND MAGNETISM **12 Hrs.**

Vector Algebra, Fundamental theorems of Gradient, Curls, and Divergence, Curvilinear co-ordinates, Continuous charge distribution, Divergence and Curl of Electrostatic Field, Electric potential and its applications, Work and Energy in Electrostatic, Bio-Savart's law, Divergence and curl of magnetic fields, Vector Potential, Ohm's law, EMF, Faraday's law of electromagnetic induction, Energy in Magnetic Fields, Maxwell's correction to ampere's law and Maxwell's equations, Poynting Vector.

UNIT II: ELECTROMAGNETIC WAVES **10 Hrs.**

Waves equation, Reflection and Transmission of waves, Polarisation, Wave equation for E and B for monochromatic plane waves, Propagation in linear media, reflection and transmission in normal and oblique incidence, Electromagnetic waves in conductors, Frequency dependence of permittivity, Waveguides, TE waves in rectangular waveguide, The Coaxial transmission line.

UNIT III: PHYSICS OF SOLIDS **10 Hrs.**

Fermi electron gas, Fermi level and surface, Energy bands, Energy Gap, Energy and band structure of conductor, insulators and semiconductors, Intrinsic semiconductors at 0K and room temperature, Intrinsic conductivity, Types of semiconductors, doping impurities, Temperature variation of carrier concentration, Electrical conductivity in semiconductors, Hall Effect, and magnetic materials

UNIT IV: OPTICS **10 Hrs.**

Nature of light waves, Fermat's principle, Coherent Sources, Interference, Two source interference, Interference in thin films, Newton's ring, Fresnel and Fraunhofer diffraction, Diffraction from single slit and double slit, Lasers, optical fibres and Holography, Applied optics: engineering measurements.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall fundamental laws and principles of electromagnetism and optics.
CO2 : Explain concepts of waves, conductivity, and semiconductor properties.
CO3 : Utilize principles to solve problems in electricity, magnetism, and optics.
CO4 : Evaluate electromagnetic phenomena and semiconductor behaviour through experimentation.
CO5 : Assess the effectiveness of engineering design for electromagnetic and optical devices.
CO6 : Design solutions for engineering challenges involving electromagnetics and semiconductors.

TEXT/REFERENCE BOOKS

1. Griffith, D. J., Introduction to Electrodynamics, Prentice Hall.
2. M. N. Avadhanulu and P G Kshirsagar, A text book of Engineering Physics, S.Chand Publications.
3. Sears and Zemansky, University physics, Pearson publications.
4. Principles of Electromagnetics, Matthew N. O. Sadiku, Oxford publications.
5. Hecht, E., Optics, Pearson Education.
6. M. A. Wahab, Solid State Physics: Structure and Properties of Materials, Narosa Publishing House Pvt. Ltd.-New Delhi.

<Course Code>					Applied Physics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Understand principles of electromagnetism and their experimental applications.
2. Analyse experimental setups and procedures related to electricity and magnetism.
3. Apply concepts of electromagnetic phenomena and optics in practical experiments.
4. Investigate semiconductor device electrical properties through experimentation.

LIST OF EXPERIMENTS

1 To determine e/m using Thomson's method.
2 To study Bio-Savart's Law.
3 To verify Faraday and Lenz's law.
4 To study the magnetic field along the axis of a coil
5 To determine the electrical conductivity of metals.
6 To study the characteristics of Si solar cells.
7 To study the phenomenon of photoconductivity using CdS photo-resistor.
8 To determine energy band gap of semiconductor using four probe method.
9 To study the hall effect and determine hall voltage, hall coefficient, type of majority charge carriers, carrier concentration and hall angle.
10 To study of the ferromagnetic hysteresis.
11 To determine the wavelength of monochromatic light (sodium light) using Newton's rings.
12 To measure the slit width of single, blade and double slits.
13 To understand fundamental of optical fibres and analogue optical fibre communication.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recognize electromagnetic principles and practical applications in experimental setup.
CO2 : Interpret experimental setups and procedures for electricity, magnetism, and optics.
CO3 : Demonstrate and implement the concepts of electromagnetic phenomena and optics.
CO4 : Investigate the electrical properties of semiconductor device.
CO5 : Examine the experimental data to identify trends and physical relationships.
CO6 : Design experiments to investigate electromagnetism and semiconductor properties.

TEXT/REFERENCE BOOKS

1. Griffith, D. J., Introduction to Electrodynamics, Prentice Hall.
2. M. N. Avadhanulu and P G Kshirsagar, A text book of Engineering Physics, S. Chand Publications.
3. Sears and Zemansky, University physics, Pearson publications.
4. Principles of Electromagnetics, Matthew N. O. Sadiku, Oxford publications.
5. Hecht, E., Optics, Pearson Education.
6. M. A. Wahab, Solid State Physics: Structure and Properties of Materials, Narosa Publishing House Pvt. Ltd.-New Delhi.

<Course Code>					ENVIRONMENTAL SCIENCE					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop a comprehensive perspective of environment and sustainable development
2. To understand the causes and effects of various types of pollution
3. To develop an understanding of the various strategies for controlling the pollution
4. To introduce the emerging environmental domains

UNIT I : INTRODUCTION TO ENVIRONMENT **05 Hrs.**

Sustainable Development; Sustainable Development Goals; Environmental Studies – Its importance and Multidisciplinary nature, Introduction to Environmental Parameters and their standards (air, water, soil, noise, etc.); Ecosystem and its types, Ideal ecosystem, Biodiversity : Its importance and conservation.

UNIT II : MULTI-SCALE ENVIRONMENTAL POLLUTION (GLOBAL, REGIONAL AND LOCAL) **06 Hrs.**

Pollution, Causes and Effects of different types of pollution : Air Pollution, Water Pollution, Soil Pollution, Solid Waste (organic and Inorganic) Pollution, Hazardous Waste Pollution, Marine Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution; Introduction to man-made disasters like floods, heat waves, landslides, etc., Introduction to the various instruments for measuring air pollution, water pollution, noise, etc.

UNIT III : ENVIRONMENTAL POLLUTION CONTROL STRATEGIES **09 Hrs.**

Multi-approaches for reducing various types of pollution: Introduction to Water and Wastewater treatment technologies, Air and Noise pollution control techniques, Introduction to different environmental management concepts like Swachh Bharat Mission, Mission LiFE (Lifestyle For Environment), etc. Indian Culture and Traditional Wisdom for managing environment

UNIT IV: EMERGING ENVIRONMENTAL MANAGEMENTS DOMAINS **08 Hrs.**

Concept of Zero Liquid Discharge (ZLD) and the reuse of the treated wastewater, Green Credit Rules - 2023, Clean Development Mechanisms (CDM) and Carbon Credits, Green Buildings, Carbon Footprint and Water Footprint, Green Business, International Environmental Laws, Environmental Auditing

TOTAL HOURS: 28 Hrs.

COURSE OUTCOMES:

On completion of the course, student will be able to:

CO-1: Demonstrate comprehension of sustainable development and environmental aspects.
CO-2: Recognize the interdisciplinary characteristics inherent in Environmental studies.
CO-3: Evaluate the impact of various pollutants on the environment.
CO-4: Assess the efficacy of different technologies for environmental pollution control.
CO-5: Analyze different environmental management policies and their implications.
CO-6: Synthesize knowledge about emerging environmental management paradigms.

TEXT-BOOK AND REFERENCE BOOKS:

1. Bharucha Erach, Textbook for Environmental Studies, UGC New Delhi.
2. Daniel B. Botkin & Edwards A. Keller, Environmental Science, Wiley India edition.
3. Miller T. G. Jr., 2006. Environmental Science, Clengage Learning.
4. R. Rajagopalan, Environmental Studies, Oxford University Press.
5. Gilbert Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, PHI.

<Course Code>					Workshop Practice					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand safety in various manufacturing processes.
2. Learn how to use various measuring tools for engineering applications.
3. Hands on training and preparation of job using wood, metal and sheet as per drawing.
4. Understand various manufacturing processes like machining, welding, soldering and 3D printing for prototypes.

LIST OF EXPERIMENTS

1 Introduction to Workshop safety, layout and identification of various materials- plastic, wood, metals-ferrous and nonferrous, rubber, glass etc.
2 Use of measuring tools for engineering applications
3 Fitting job: Detailed drawing of work piece, use of fitting tools and job preparation.
4 Carpentry job: Detailed drawing of work piece, use of carpentry tools and job preparation.
5 Sheet metal job: Detailed drawing of work piece, use of sheet metal working tools and job preparation.
6 Plumbing job: Internal/External threading, piping network using Tees, Elbows, Reducer, Bends etc
7 Hands on training on mini lathe and milling machine
8 Demonstration of welding, brazing and soldering
9 Soldering and desoldering for PCB
10 3D printing using polymer and metal.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define workshop safety and various engineering materials
CO2 : Understand various measuring equipment
CO3 : Apply various workshop tools in preparing job for carpentry, fitting, sheet metal and plumbing
CO4 : Examine various manufacturing operations like welding and machining
CO5 : Evaluate soldering operation for PCB
CO6 : Create prototype using 3D printing

TEXT/REFERENCE BOOKS

1. S. K. Hajra Choudhury, Elements of Workshop Technology, Vol. I & II, Media Promoters and Publishers.
2. H. S. Bawa, Workshop Practice, Tata-McGraw Hill.
3. Kalpakjian S. And Steven S. Schmid, Manufacturing Engineering and Technology, Pearson Education India Edition.
4. Rao P.N., Manufacturing Technology, Vol. I and Vol. II, Tata McGraw Hill House.

<Course Code>					Biology For Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the basic biological concepts and their engineering applications.
2. To introduce the students with an understanding of biodesign principles to create novel devices and structures.
3. To provide the students an appreciation of how biological systems can be re-designed as substitute products for natural systems.
4. To study the development of interdisciplinary vision of biological engineering.

UNIT I: INTRODUCTION TO BIOMOLECULES **7 Hrs.**

Cell as a basic unit of life, prokaryotic and eukaryotic cells, microbes, plant and animal cells; Cell organelles – structure and function; Cell membrane. Vitamins and Minerals, Carbohydrates, Lipids, Amino Acids and proteins, Nucleic Acids (DNA and RNA).

UNIT II: NATURE-BIOINSPIRED MECHANISMS **7 Hrs.**

Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics). The eye functions as a camera. Heart as a pump system, The ear and nose as signal transmission models. Lungs operate as an exchange model, kidneys function as a filtration system. Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro).

UNIT III: TRENDS IN BIOENGINEERING **7 Hrs.**

Stem cell and its applications, Bio printing techniques and materials. Applications of Bioinformatics, Artificial Intelligence for diagnosis and detection of communicable and non-communicable diseases, Biosensors in healthcare, Environmental monitoring, Food Safety and Biosecurity.

UNIT IV: ENGINEERING PERSPECTIVES OF BIOLOGICAL SCIENCES **7 Hrs.**

Biology and engineering crosstalk – At cell level: Hybridoma technology, At tissue level: Plant Tissue Culture, Animal Tissue Culture; Tissue Engineering, Introduction to Biomimetics and Biomimicry, Nanobiotechnology, Introduction to Radiology, High-throughput diagnostics in clinics: Molecular Diagnostics (PCR), DNA chips.

TOTAL HOURS: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain fundamental knowledge of origin of life.
CO2 : Demonstrate the involvement of major organs and systems in the human body as models for bioengineering design.
CO3 : Analyse the scope and opportunities of Implementing bioengineering concepts relates to advanced imaging and diagnostic techniques for clinical diagnosis and disease management.
CO4 : Get acquainted with concepts of bioengineering trends including biosensors, AI , Imaging techniques in disease diagnosis and treatment
CO5 : Focus on implications of Bioinspired designs, Bioengineering trends including, tissue engineering and molecular diagnostics.
CO6 : Think critically towards exploring innovative biobased solutions for socially relevant problems.

TEXT/REFERENCE BOOKS

1. Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M., Biology for Engineers, Tata McGraw-Hill, New Delhi.
2. Arthur T. Johnson, Biology for Engineers, CRC Press, Taylor and Francis.
3. Geoffrey M.Cooper, The Cell: A molecular Approach, ASM Press.
4. Sohini Singh and Tanu Allen, Biology for Engineers", Vayu Education of India, New Delhi.
5. Yoseph Bar-Cohen, "Biomimetics: Nature-Based Innovation, CRC Press.
6. Stuart Fox, Krista Rompolski, Human Physiology", McGraw-Hill (eBook).
7. D. Floreano and C. Mattiussi, Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies, MIT Press.

<Course Code>					Computer Programming – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To understand the usage of operators and data types.
2. To apply different types of Conditional and looping statement.
3. To create different types of data collections.
4. To implement user defined function.
5. To perform different operations upon files.

UNIT 1: Basics of Programming **3 Hrs.**

Introduction to Computer Programming, Features of C language, Structure of C program, program execution flow, C Tokens, variables, Data types, Operators, Decision control statements-if, switch, go to statement. Loop control structures- while, do-while, for loop, Break statement, Continue statement

UNIT 2: Derived Data types **4 Hrs.**

Array: one dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, String-Basic Concepts, Inbuilt String manipulation Functions, Pointer, Pointer arithmetic, Pointer to pointer, Array of Pointers

UNIT 3: Functions and Structures **4 Hrs.**

Introduction to user defined functions, Types of Functions, Call by value-call by reference, recursion, pointers to functions, Structures, Array of Structure, Union

UNIT 4: Files Handling **3 Hrs.**

File handling in C, Different types of files, Operations on Files such as File creation, File deletion, File access modes such as read, write, append, File concatenation, File handling using seek function.

14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Understand basics of programming.
CO2: Identify different programming constructs for a problem.
CO3: Apply appropriate derived data type for data storage.
CO4: Prepare a user defined data type based on data attributes.
CO5: Construct user defined functions for problem solving.
CO6: Analyse different data structure based on application requirement.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language, PHI
2. K. N. King , C Programming: A Modern Approach, W.W. Nortorn
3. David Griffiths and Dawn Griffiths, Head First C: A Brain-Friendly Guide, O'Reilly
4. E.Balaguruswamy, Programming in ANSI C, McGraw-Hill
5. Y.P. Kanetkar, Let us C, BPB Publication
6. Y.P. Kanetkar , Pointers in C, BPB Publications

<Course Code>					Computer Programming – I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To implement basic programming concepts.
2. To create different types of data collections.
3. To implement user defined function.
4. To perform different file handling operations.

List of Experiments:

1. **Introduction to Computer Programming**: Understanding compilation process through a simple C program, program execution flow, C Tokens, variables and keywords and identifiers, types of C constants and variables.

2. **Simple and formatted Input Output Operations:** Data types, Operators, Input /output statements in C, Formatted I/O, format specifiers, escaper sequences
3. **Decision making and branching:** if, if-else, if-else ladder, switch, go to statement, conditional operator statement

4. **Looping control structures:** while, do-while, for loop, Break statement, Continue statement
5. **Derived Data Type: Array and Strings:** One dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, Basic Concepts, Inbuilt String manipulation with and without using inbuilt functions.
6. **Derived Data Type: Structure and Union:** structure, arrays and structures, structures and functions, pointer to structure, typedef, unions
7. **Functions:** Introduction to user defined functions, Types of Functions, Call by value-call by reference, header file creation, recursion, pointers to functions, arrays and functions
8. **Pointers:** Pointer's basics, use of &, * operator in context to pointers, Pointer arithmetic, Array and String processing using pointer, pointer to pointer, Array of Pointers
9. **File Handling in C:** File handling in C, Different types of files, Operations on Files, File handling functions.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Identify the use of appropriate naming conventions and programming style including appropriate comment density.
CO2: Implement a basic C program using appropriate control structure.
CO3: Apply appropriate derived data types based on data attributes.
CO4: Develop solutions with pointers and utilize them to access strings and structures.
CO5: Design user defined functions for problem solving and reuse them across different programs.
CO6: Apply suitable file handling functions and operations.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language, PHI.
2. K. N. King , C Programming: A Modern Approach, W.W. Norton.
3. E. Balaguruswamy, Programming in ANSI C, McGraw-Hill.
4. Y.P. Kanetkar,Let us C, BPB Publication.
5. Y.P. Kanetkar, Pointers in C, BPB Publication.

<Course Code>					Universal Human Values					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the need of nurturing human values through the process in value-based education system.
2. To understand and develop a holistic perspective on self-exploration and being in harmony with family, society and nature.
3. To facilitate the students in understanding harmony at all the levels and applying in their profession and work place to lead an ethical life.

UNIT I: HUMAN VALUES AND PROCESS OF VALUE EDUCATION **04 Hrs.**

Human values, human aspirations and the ultimate goal, understanding happiness and prosperity, appraise the meaning of satisfaction and happiness in current scenario, harmony and compatibility, values imbibed education system and process

UNIT II: KNOWING SELF - HARMONY WITH SELF **04 Hrs.**

Understanding self, capabilities and challenges, understanding material (physical facilities) and spiritual needs - need of mind and body, understanding body as an instrument, harmony between mind and body, synchronizing physical health and mental health, practicing healthy habits for healthier me

UNIT III: HARMONY IN RELATIONSHIP – FAMILY, SOCIETY AND NATURE **03 Hrs.**

Harmony in relationships, values for harmony in any human-human interaction, harmony in family, and society, trust and respect for others, self esteem and ego, equality, equity, inclusion and liberation, concept of '*Vasudhaiva Kutumbakam',* understanding co-existence and sync with nature

UNIT IV: HARMONY IN PROFESSION AND ETHICAL BEHAVIOR **03 Hrs.**

Ethical human conduct, acceptance and respect, appraising the qualities of others, professional competence for enabling harmony in system and enabling universal human order, scope of eco-friendly systems, strategies to reach the harmonious ecosystem to reach Universal Human Order '*Sarvabhauma Vyavastha'*

TOTAL HOURS: 14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 - Understand the significance of human values, its need, and process of value education.
- CO2 - Appraise the meaning of happiness and prosperity as short- and long-term goal of life. Understand them and in context of the current scenario
- CO3 - Distinguish between the mind and body, physical and spiritual wellbeing for harmony within self
- CO4 - Assess the value of harmonious relationship based on trust, respect and enduring its role in all human-human relationships to build harmonious society
- CO5 - Understand the importance of harmony with nature and appreciate co-existence for harmonious ecosystem.
- CO6 - Create the perfect professional place and work environment following the ethical practices and strategize to uphold the human values at all the levels and interactions.

TEXT/REFERENCE BOOKS

1. R. R. Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and Professional Ethics, Excel books.
2. A. Nagraj, Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
3. A. N. Tripathi, Human Values, New Age Intl. Publishers.
4. M. K. Gandhi. The Story of My Experiments with Truth, Fingerprint Publishing.
5. Ivan Illich, Energy & Equity, The Trinity Press, Worcester, and Harper Collins.
6. E. F. Schumacher, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.
7. Sussan George, How the Other Half Dies, Penguin Press.